

Subhatra Sivam

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Earth System Scientist

Education

University of California, Santa Barbara (UCSB)

Earth Science, Ph.D.

June 2030

University of Maryland, College Park (UMD)

Overall GPA: 3.88/4.0

Atmospheric and Oceanic Sciences, B.S. (High Honors)

May 2023

Theatre, B.A.

May 2023

Minors: Archaeology, Surficial Geology

Undergraduate Thesis: *Comparing Air-Sea Heat Flux over the Pacific Arctic Ocean from Saildrone Observations and Global Flux Products*

Work/Research Experience

Los Alamos National Lab Intern

July 2026 – September 2026

Carr Astronautics Atmospheric Scientist

June 2023 – July 2025

- ◆ NOAA/NESDIS/STAR: Research remote sensing uses and needs for NOAA/NESDIS operations.
 - Reconstructed radiance data by using hybrid principal component analysis on operational Cross-Track Infrared Sounder (CrIS) data and checked its effect on numerical weather prediction models.
 - Helped integrate Infrared Atmospheric Sounding Interferometer (IASI) with CrIS data to increase satellite imagery coverage.
 - Researched space weather sensors, variables, and data characteristics to assess efficiency and cost-effectiveness of sensors used in NOAA space weather operations.
 - Assess impact of NOAA space weather data and research on space traffic management sector.
- ◆ NASA: Developed regression model relating light types and nighttime hyperspectral radiance satellite data.

NOAA Lapenta Intern

June 2022 – May 2023

- ◆ Compared air-sea heat flux between three global reanalysis products (MERRA2, ERA5, CFSv2) and saildrone observations in the Bering, Chukchi, and Beaufort Seas.

UMD Environmental Monitoring Lab Researcher

September 2020 – May 2021

- ◆ Monitored carbon dioxide levels in and around the UMD Campus through low-cost sensors.

John Hopkins University Applied Physics Lab ASPIRE/College Intern

September 2018 – March 2022

- ◆ Integrated artificial light behavior and propeller wakes into a MATLAB-based ocean simulation.
- ◆ Developed a python-based interface for real-time data acquisition in hydrodynamics testing facility.
- ◆ Created a python-based interface for a Bayesian network used to predict underwater munition locations in an old testing site.
- ◆ Compared radiative transfer calculations between two MATLAB-based ocean simulations.

Howard County Conservancy Intern

September 2017 - June 2018

- ◆ Compiled data about the environmental quality of Howard County schoolyards and local tributaries.
- ◆ Worked with high school students to run riparian experiments.

Publications

Sivam S, Zhang C, Zhang D, Yu L and Dressel I (2024) Surface latent and sensible heat fluxes over the Pacific Sub-Arctic Ocean from saildrone observations and three global reanalysis products. *Front. Mar. Sci.* 11:1431718. doi: 10.3389/fmars.2024.1431718

Posters

- ◆ Sivam S., Matoza R.S. Attribution of Background Seismoacoustic Signals to Coastal Surf. *University Program Review 2026.*
- ◆ Sivam S., Matoza R.S. Seismic and Infrasonic Signals in Icy Environments. *Acoustical Society of America 2026.*
- ◆ Sivam S., Matoza R.S. Cryospheric Clutter: Using Collocated Seismic and Infrasound Sensors to Quantify Surf in Coastal Alaska. *Seismological Society of America 2026.*
- ◆ Sivam S., Zhang C. Surface latent and sensible heat fluxes over the Pacific Arctic Ocean from saildrone observations and three global reanalysis products. *American Meteorological Society 2024.*

Teaching Experience

UCSB Teaching Assistant

- ◆ Principals of Physical Geology (Earth 2) *September 2025 – December 2025*

UMD Undergraduate Teaching Assistant

- ◆ Large-Scale Dynamics of the Atmosphere and Ocean (AOSC432) *January 2023 – May 2023*
- ◆ Atmospheric Thermodynamics (AOSC431) *September 2022 – December 2023*

Honors, Awards, and Certificates

Nuclear Security and Nonproliferation Summer School

June 2026

- ◆ Certificate of completion awarded by the Nuclear Science and Security Consortium and the Lawrence Livermore National Laboratory.

Physical Acoustics Summer School

May 2026

- ◆ Certificate of completion awarded by the National Center for Physics and the University of Mississippi.

Mildred E. Mathias Graduate Student Research Grant

March 2026

- ◆ Grant awarded by the University of California Natural Reserve System.

Henry Fleming Award for Outstanding Senior

May 2023

- ◆ Awarded by the UMD Department of Atmospheric and Oceanic Science.

Science Diplomacy Citation

December 2022

- ◆ Awarded by the UMD Global Fellows Program.

Louis Allen Award for Outstanding Rising Senior

May 2022

- ◆ Awarded by the UMD Department of Atmospheric and Oceanic Science.

Honors Citation

December 2021

- ◆ Granted by the Honors College: University Honors Program.

Robert Brammer Undergraduate Prize Competition in Climate Finance – Second Place. *May 2021*

- ◆ Awarded by the UMD Department of Atmospheric and Oceanic Science and Smith School of Business.

Skills

Programming: MATLAB, Python, Linux, HTML/CSS, Fortran, LaTeX

Field Research Techniques: photogrammetry

Audiovisual Technology: light board operation and programming, live sound mixing, microphones

Audiovisual Software: QLab, Logic Pro, GarageBand, Adobe Premiere, Adobe Photoshop, Adobe Audition

Language: English, Tamil, Spanish